

Claims and limitations

Repository source: certificate/CLAIMS_AND_LIMITATIONS.md

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Main claim represented by the package

The exact certificate and manuscript represent the universal statement

$$\text{Vol}(K) \geq \frac{26220940089713\pi}{20000000000000} d^3 = 0.41187756378030227424817892598 \dots d^3$$

for every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$.

The claim is universal: it is not restricted by symmetry, smoothness, contact type, prescribed circumradius, or proximity to a Meissner body. Scaling from width one supplies the factor d^3 .

Current verification classification

Proposed exact universal certificate - standalone internal replay.

This means:

- the finite certificate is represented by exact integer and rational objects;
- the standalone exact verifier accepts those objects;
- the assembled theorem certificate is reconstructed byte for byte;
- independent binary64 Python and base-R audits agree;
- ten reconstruction/regression tests pass;
- seven targeted corruptions are rejected;
- paper-facing constants and the lower-row table are generated or reconciled against the certificate JSON; and
- the package is still an AI-assisted, non-peer-reviewed research draft.

The word **exact** refers to the arithmetic of the finite certificate. It does not mean that the full proof has been formalized in Lean or another proof assistant, and it does not mean that the argument has been independently reviewed.

Claims established inside the finite package

Subject to the mathematical reductions in the manuscript, the exact package checks:

- gap-free circumradius coverage from $1/2$ through $\sqrt{6}/4$;
- the exact split $R_* = 305695601/500000000$;
- the unique lower-branch bottleneck 03B02;

- four exact replacement cells for the old 03A1 interval;
- reconstruction of a 98,306-vertex near-Jung fan;
- all exact fan-margin and inherited-multiplier inequalities;
- the determinant transfer and forced-core lower bound;
- the dominant-edge coordinate-width estimate on all three axes;
- three optimized antipodal cap-pair contributions;
- exact six-cap disjointness;
- a strict near-Jung handoff surplus; and
- the exact theorem coefficient and decimal enclosure.

What is not claimed

The package does **not** establish any of the following:

- that the bound is sharp;
- an equality case;
- rigidity or stability near equality;
- the identity or uniqueness of a minimizing body;
- that either Meissner body minimizes volume;
- the Meissner conjecture;
- a classification of all extremal contact configurations;
- a proof in Lean, Coq, Isabelle, or another proof assistant;
- external independent reproduction;
- peer review, journal acceptance, or publication;
- a systematic priority or novelty guarantee; or
- correctness of every statement merely because it was generated or checked by an AI system.

The explicit Meissner value

$$\frac{\pi}{12} \left(8 - 3\sqrt{3} \arccos \frac{1}{3} \right) d^3 = 0.419860045965080 \dots d^3$$

is used only as an explicit-body comparison value. It is not asserted here as the universal minimum.

Relationship to earlier bounds

The certificate coefficient exceeds, by direct arithmetic comparison, the public HYRA coefficient

$$\frac{130838246407123\pi}{10^{15}} > 0.411040473721188.$$

It also exceeds Nishioka's $4\pi/33$ coefficient and Chakerian's historical coefficient. These comparisons do not by themselves establish priority or publication status. The dated source review is recorded in [../proof/SOURCE_LEDGER.md](#).

Smoothness and approximation

The final theorem is stated for all convex bodies of constant width, including nonsmooth bodies. The analytic divergence step is first justified on a regular class and then passed to arbitrary bodies by the regularization and continuity argument stated in the manuscript. The finite verifier assumes that reduction; it does not independently formalize geometric measure theory or Hausdorff continuity.

Status words that may be used

For public release and before external review, acceptable descriptions include:

- proposed exact universal certificate;
- computer-assisted research draft;
- exact-rational finite verification;
- independently audited in Python and base R; and

- seeking independent mathematical verification.

Descriptions that should not be used without new evidence include:

- peer reviewed;
- independently verified;
- formally verified;
- sharp or optimal;
- proof of the Meissner conjecture;
- identification of the minimizer; and
- final accepted theorem.

Corrections policy

A mathematical or certificate correction must be visible. It should receive a new commit and, after release, a new version or release tag. The old tagged artifact should remain available with a clear correction notice rather than being silently rewritten.

A change to the exact theorem coefficient, branch partition, proof objects, or verifier semantics requires:

1. a new certificate schema or documented schema revision;
2. regenerated exact and audit transcripts;
3. renewed adversarial tests;
4. paper and documentation reconciliation;
5. fresh-machine replay; and
6. explicit release notes comparing the old and new claims.